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Crown Scorch Height

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Synonyms

Canopy scorch height; Crown scorching; Height of crown scorch; Height of lethal scorch; Height of lethal scorching; Height of scorch; Lethal scorch height; Scorch height; Scorching

Definition

Crown scorch height represents the vertical distance above the ground that lethal scorching of the needles or leaves in the canopy of a forest or shrubland ecosystem has occurred as a result of the heat generated from a surface fire.

Introduction

The hot, rising convective gases and radiant heat from the combustion zone of a surface fire can possibly kill the overstory foliage in a forest or brushfield without consuming the needles or leaves. This vertical dimension of lethal scorching, as typically reflected in a distinct color change, is referred to as the crown scorch height (Fig. 1). Related metrics include crown scorch volume and length expressed as a percentage of the pre-fire values (Peterson and Ryan 1986; Hood 2007b). Note that crown scorch height as used here focuses solely on foliage damage of conifer trees, although the general principles apply to all woody plants (e.g., tall shrubs). It does not address bud or limb cambial damage (Peterson and Ryan 1986; Wade and Johansen 1986; Michaletz and Johnson 2006a; Hood 2007a) and the implications for tree mortality (Michaletz and Johnson 2007).

The full impacts of wildfire or prescribed fire on wildland ecosystems are not always immediately evident. Crown scorch height is a good case in point. Soon after a fire, well-scorched foliage of some species may show a color change, while lightly scorched foliage may have a dull appearance, turning to yellow and brown over a period of days to weeks and brownish red over several months (Fig. 1). When the crown scorch does become evident, it can usually be recorded as a distinct height above the ground surface. Below the crown scorch line, all the needles are discolored and dead (Fig. 2a) or possibly consumed by flames (Fig. 2b) and above it, green and alive; the differences between these two situations can to a certain extent be quantified as illustrated in Fig. 3 (Alexander and Cruz 2012c). For an excellent set

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Crown Scorch Height, Fig. 1 Crown scorching of a mature ponderosa pine (*Pinus ponderosa*), as evidenced by the brownish red color of the needle foliage in the lower crown, following a low-intensity prescribed fire in the Dewdrop Range near the community of Kamloops in South-Central British Columbia, Canada, on May 18, 1978. Photo taken September 26, 1978 by John V. Parminter

of color photos depicting crown scorch height, see Dieterich (1979).

The degree of heat scorch and specific plant characteristics, such as size (e.g., tree height), dictate in part the likelihood of tree mortality in many species (Fernandes et al. 2008; Woolley et al. 2012), although other factors like growing season or time of year, for example, play a role (Wade and Johansen 1986; Burrows 1997). Thus, crown scorch height is an important consideration in developing a prescribed burn prescription and also a criterion for salvage logging following fire in many forest types (Peterson and Ryan 1986; Hood et al. 2007).

Theory

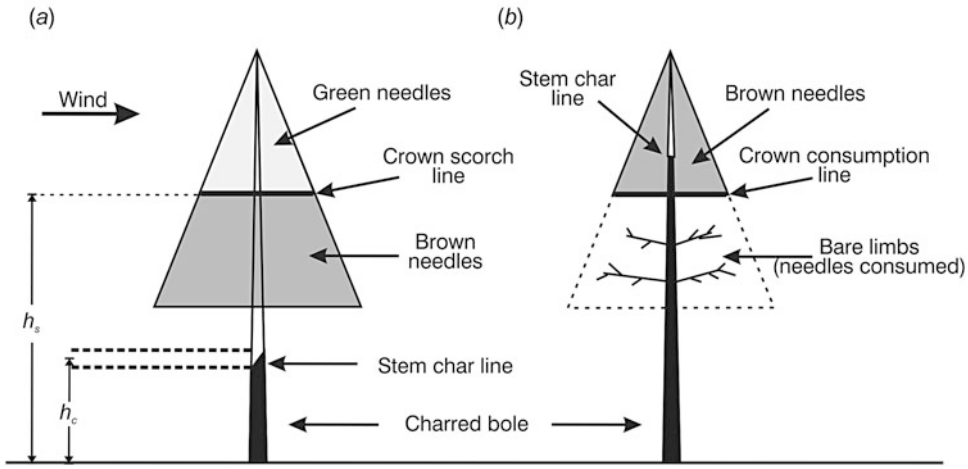
The temperature reached at a given height in the convection column or plume above a surface fire in a forest depends on the energy release rate, as quantified by fireline intensity (Byram 1959), ambient air temperature, and the in-stand wind speed. The maximum crown scorch height represents the point above the ground surface where an air temperature (assumed to be equivalent to the needle or leaf temperature) of 60 °C is reached, which is generally regarded as the lethal temperature for crown foliage (Byram 1958), although, in fact, a time-temperature relationship exists as opposed to a distinct temperature threshold (Alexander 1998). For a lethal temperature of 60 °C, the needle exposure time is considered to be around 30 s (cf. Nelson 1952). This is roughly the nominal flame-front residence time for surface head fires in many forest types (e.g., Lawson 1972; Wotton et al. 2012).

Van Wagner's Equations

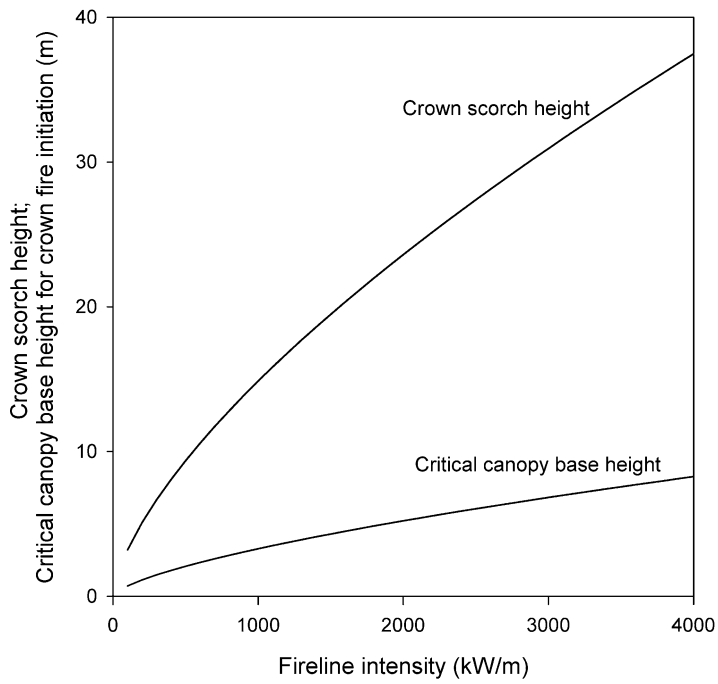
Van Wagner (1973) derived an empirical relation between Byram's (1959) fireline intensity (I_B, kW/m) and crown scorch height (h_s, m) (after Alexander 1982):

$$h_s = 0.1483\ I_B^{0.667} \tag{1}$$

Equation 1 was developed with data from 13 experimental fires carried out in Eastern Ontario, Canada, with I_B and h_s varying from 67 to 1255 kW/m and 2 to 17 m, respectively. Eight of the fires were in a stand comprised of red pine (*Pinus resinosa*) and eastern white pine (*P. strobus*), two in jack pine (*P. banksiana*), one in northern red oak (*Quercus rubra*), and the remaining two in a red pine plantation. The experimental fires were carried out under moderate to high ambient air temperatures (23–31.5 °C) and light in-stand winds (2.3–4.7 km/h). All of the fires were conducted on flat ground. The results agreed with established theory that h_s varies with the 2/3 power of I_B (Thomas 1963).



Crown Scorch Height, Fig. 2 Variation in direct effects of (**a**) low- to (**b**) high-intensity surface fires on tree crowns and stem boles as illustrated by Storey and Merkel (1960) as observed on the Buckhead wildfire of March 1956 in North Florida, USA (From Alexander and Cruz 2012a). Symbols: h_s, crown scorch height and h_c, stem-bark char height



Crown Scorch Height, Fig. 3 Graphical representation of the relationships between crown scorch height based on Van Wagner's (1973) model and partial canopy foliage consumption based on Van Wagner's (1977) model of the critical surface intensity for crown combustion, assuming a foliar moisture content of 100%, both as a function of Byram's (1959) fireline intensity

As the rate of spread of a fire varies around its perimeter (Catchpole et al. 1992), so too does I_B. Thus, h_s varies in a similar manner given by the relationship to I_B as stipulated by Eq. 1.

As depicted in Fig. 2, h_s and stem-bark char height are two physical, post-fire indicators of fire behavior. Several investigators have, for example, used the inverse of Eq. 1 (Alexander and Cruz 2012a, b) to obtain a rough approximation of

I_B from field measurements of h_s in the absence of on-site observation of a fire's behavior. Trees are however not perfect sensors. The limitations of this method include h_s exceeding the height of small trees and not reaching the crown base height of tall trees (Martinson and Omi 2008).

Van Wagner (1973) also developed separate relations that accounted for ambient air temperature (T_a, °C), lethal temperature (T_L, °C), and in-stand wind speed measured at a height of 1.2 m above ground (U, km/h) as additional controlling variables (after Alexander 1985):

$$h_s = \frac{4.4713\ I_B^{0.667}}{T_L - T_a} \tag{2}$$

$$h_s = \frac{0.74183\ I_B^{7/6}}{(0.025574\ I_B + 0.021433\ U^3)^{0.5}\ (T_L - T_a)} \tag{3}$$

Van Wagner (1973) specified that $T_L = 60$ °C.

Equations 1, 2, and 3 were parameterized by fitting a linear model through the origin to transformed data. The models produced an r^2 of 0.98. Van Wagner (1973) was to note that such a degree of fit was probably somewhat fortuitous in view of the rough nature of the data.

Within the US wildland fire management community (Hood 2007c), the corresponding units for I_B, h_s, U, and both T_L and T_a are Btu/(s-ft), ft., mi/h, and °F respectively. For a list of conversion factors, see Van Wagner (1978).

Other Model Relationships Between Fireline Intensity and Crown Scorch Height

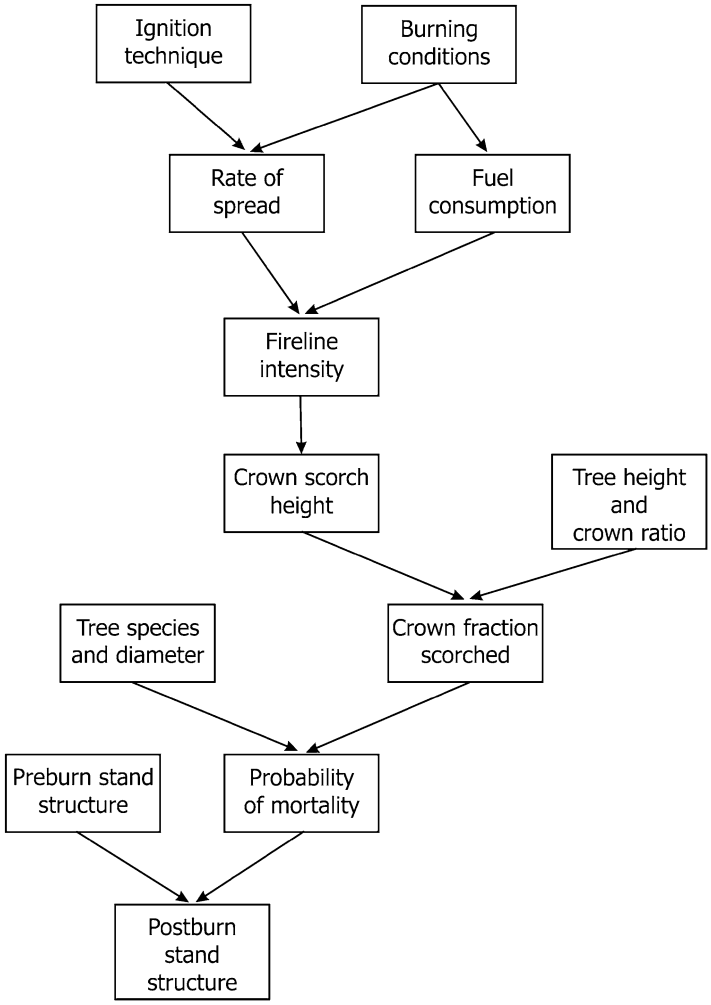
Van Wagner (1973) was the first to formally derive and publish an empirical relation between I_B and h_s. As a result, there has been an overwhelming tendency within the wildland fire community to regard Van Wagner's (1973) equations as universal in nature (Alexander and Cruz 2012a). For example, a list of 11 other $h_s - I_B$ equations, including the experimental data range in the variables associated with these equations, is presented in Alexander and Cruz (2012a). Seven of the 11 equations also employed the 2/3 power of I_B.

Graphical representation of the 12 $h_s - I_B$ equations (Alexander and Cruz 2012a, b) shows considerable variation among relationships. This is believed to be due to the differences in tree species, stand structure, surface and ladder (e.g., bark) fuel characteristics, measurement methodologies, type of ignition or firing pattern, and the combustion environment (Alexander and Cruz 2012a).

Modelling Systems

Several fire modelling systems developed in the USA have incorporated Eq. 3 in order to allow users to estimate h_s. These include BehavePlus (Andrews et al. 2008), First Order Fire Effects Model (FOFEM) (Reinhardt et al. 1997), and the Fire and Fuels Extension to the Forest Vegetation Simulator (FFE-FVS) model (Rebain 2015). Similar implementations have also occurred in Europe (e.g., Fernandes et al. 2012) and Canada (Taylor and Armitage 1996; Michaletz and Johnson 2006b). All of these software programs, except for one, assume a single, heading line-source ignition pattern or firing technique. The Taylor and Armitage (1996) SCORCH program includes provision for point-source fires and different line-fire source spacing (Fig. 4). Due to the inherent temporal and spatial variations in fuels, weather, and topography in a good many fire environments, users of these fire modelling systems should in practice not expect a high degree of predictive accuracy as illustrated, for example, by Fig. 5.

None of the h_s-I_B models developed to date are suitable for cases of sloping terrain. They would all tend to underpredict h_s (Rothermel 1985). However, higher than expected scorch heights can be expected to occur at ridge lines (Rothermel 1985).



Crown Scorch Height, Fig. 4 Flow chart illustrating the inputs and processes associated with the SCORCH program. (After Taylor and Armitage 1996)

Relevance to Wildfires and Wildland-Urban Interface Fires

Crown scorch of the overstory is a significant outcome of both prescribed fires and wildfires burning beneath forest canopies within and outside the wildland-urban interface (Scott et al. 2014). Maximum crown scorch heights are often used to define a burning prescription as mentioned earlier, with predicted crown scorch height being compared to stand characteristics to determine whether a burning prescription is acceptable or not (Reinhardt and Ryan 1988; de Ronde et al. 1990; Fernandes et al. 2012). The degree of crown scorching is often used as the primary indicator of prescribed burn success in certain forest ecosystems (Wade and Lunsford 1988). In the case of wildfires, postburn h_s assessments are utilized in judging fuel treatment effectiveness (Cram et al. 2006; Martinson and Omi 2008; Safford et al. 2012), and simulated values are useful in long-term fire and fuel management planning (Salazar and Bradshaw 1986; Stephens 1998).

Crown Scorch Height, Fig. 5 Variation in crown scorch heights in a naturally regenerated maritime pine (*Pinus pinaster*) stand following a wildfire in the Alvão Mountains of Northern Portugal on August 4, 2005. Photo taken August 26, 2005 by Paulo M. Fernandes

Cross-References

- ▶ Crown Fire
- ▶ Fireline Intensity
- ▶ First-Order Fire Effects Model (FOFEM)
- ▶ Flame Front Dimensions
- ▶ Surface Fire

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